**SIMATS ENGINEERING**

**ASSESSMENT TOOL 2: Scenario-Based Questions**

**COURSE NAME: COMPUTER VISION** **COURSE CODE: ITA05**

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**COURSE OUTCOME COVERED**

**CO2:** *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

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**QUESTIONS: 5** **Total Marks50** **Duration :60 Minutes**

**Annexure A**

**Scenario-Based Questions**

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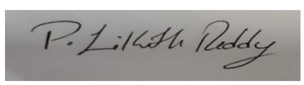
***Code of Conduct:***

*I P LIKITH REDDY (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** 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**RUBRICS FOR CALCULATION:**

| **Criteria** | **Weightage** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** | **Marks (100)** |
|---|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario | |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues | |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts | |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided | |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand | |

**Scenario-Based Questions**

**1**. Object Boundary Detection
In a medical image, boundaries between tissues need to be clearly identified.
(a) Interpret the scenario and explain the importance of boundary detection.
(b) Identify key challenges in detecting boundaries.
(c) Apply a suitable edge detection method.
(d) Justify your selection with logical reasoning.
(e) Present your explanation clearly and systematically.

ANSWER:

**1. Object Boundary Detection**

**(a) Interpret the scenario and explain the importance of boundary detection**

In a medical image, different tissues (such as organs, tumors, or bones) must be accurately separated. Boundary detection identifies the edges between adjacent tissues, helping doctors locate abnormalities, measure organ size, and support diagnosis and treatment planning.

**(b) Key challenges in detecting boundaries**

- Image noise causing false edges.
- Low contrast between neighboring tissues.
- Blurred or weak boundaries.
- Presence of overlapping structures.

**(c) Suitable edge detection method**

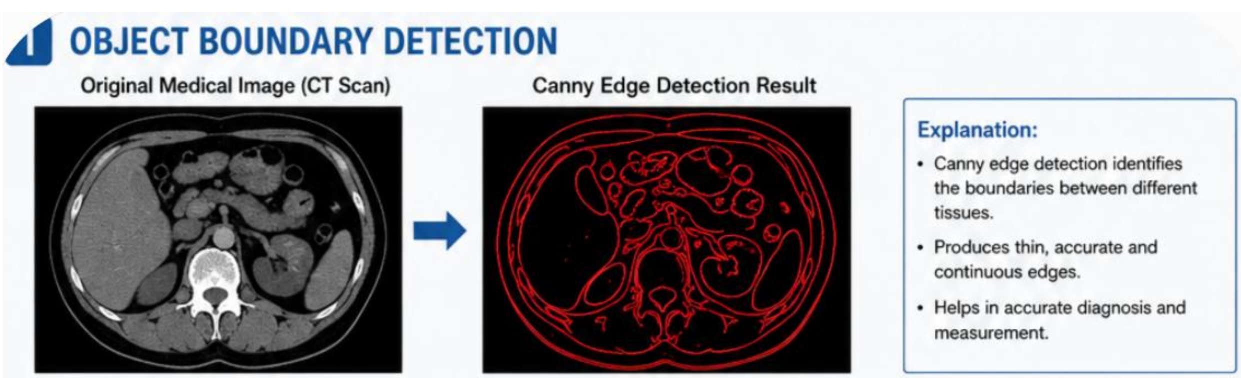
Canny Edge Detection

**(d) Justification**

- Effectively removes noise using Gaussian smoothing.
- Detects both strong and weak edges.
- Uses non-maximum suppression for thin edges.
- Employs hysteresis thresholding to produce continuous boundaries.

**(e) Structured Explanation**

- **Scenario:** Medical tissue boundary detection.
- **Problem:** Weak and noisy edges.
- **Method:** Canny Edge Detection.
- **Result:** Accurate, continuous, and reliable tissue boundaries.



2. Simple Segmentation
An image contains objects clearly distinguishable from the background based on intensity.
(a) Interpret the scenario and explain segmentation requirements.
(b) Identify key features enabling separation.
(c) Apply a suitable segmentation technique.
(d) Justify your decision logically.
(e) Present your response clearly.

ANSWER:

**2. Simple Segmentation**

**(a) Interpret the scenario and explain segmentation requirements**

The image contains objects that have intensity values clearly different from the background. The goal is to separate the object from the background accurately.

**(b) Key features enabling separation**

- High intensity difference.
- Uniform background.
- Good image contrast.
- Minimal overlap between object and background intensities.

**(c) Suitable segmentation technique**

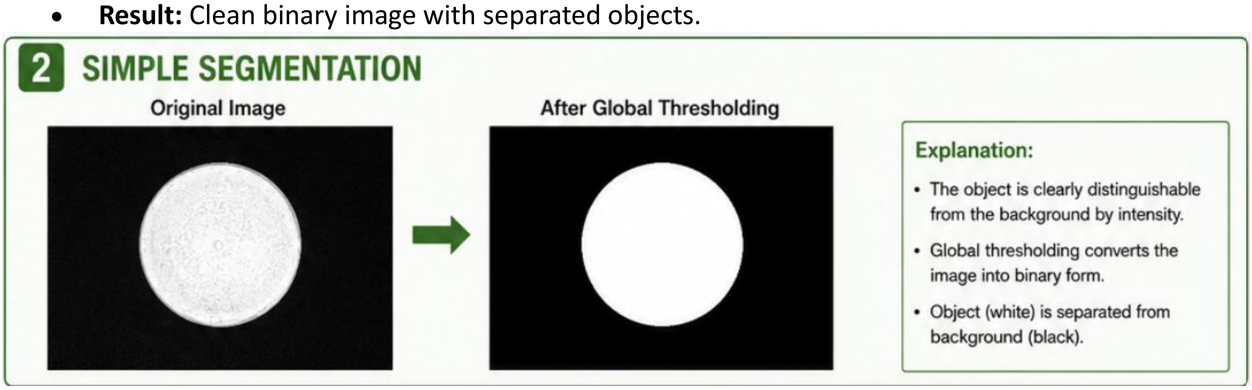
Global Thresholding (Binary Thresholding)

**(d) Justification**

- Simple and computationally efficient.
- Works well when object and background have distinct intensity values.
- Produces fast and accurate segmentation.

**(e) Structured Explanation**

- **Scenario:** Clear object-background separation.
- **Problem:** Segment object from background.
- **Method:** Global Thresholding.
- **Result:** Clean binary image with separated objects.



3. Complex Segmentation
An image contains multiple regions with overlapping intensity values.
(a) Interpret the scenario and explain segmentation challenges.
(b) Identify key issues such as intensity overlap.
(c) Apply an appropriate advanced segmentation method.
(d) Justify how the method handles complexity.
(e) Provide a structured and clear explanation.

ANSWERS:

**3. Complex Segmentation**

**(a) Interpret the scenario and explain segmentation challenges**

The image contains multiple regions with similar or overlapping intensity values, making simple thresholding ineffective.

**(b) Key issues**

- Intensity overlap.
- Image noise.
- Irregular object shapes.
- Weak boundaries between neighboring regions.

**(c) Appropriate advanced segmentation method**

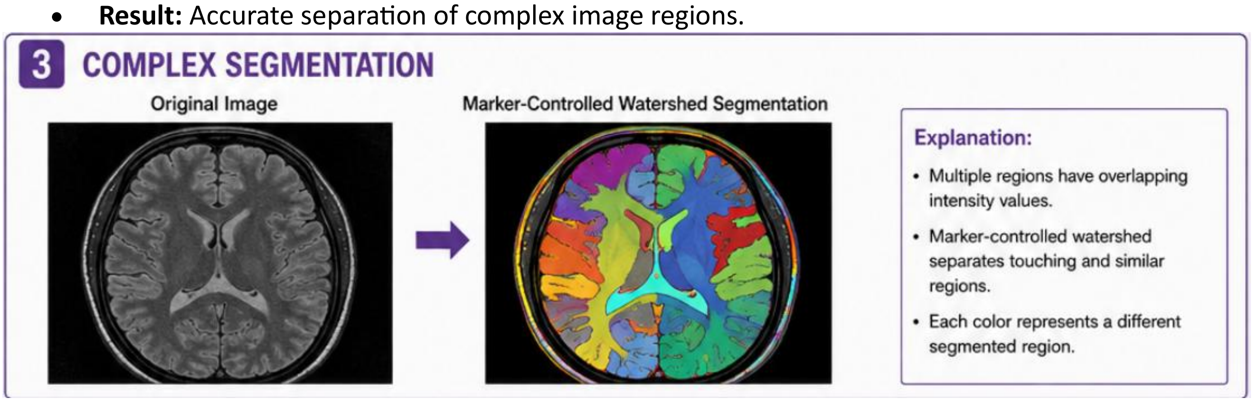
Watershed Segmentation (Marker-Controlled Watershed)

**(d) Justification**

- Separates touching and overlapping regions.
- Uses markers to reduce over-segmentation.
- Produces accurate region boundaries even with similar intensities.

**(e) Structured Explanation**

- **Scenario:** Multiple overlapping regions.
- **Problem:** Intensity overlap.
- **Method:** Marker-Controlled Watershed Segmentation.
- **Result:** Accurate separation of complex image regions.



4. Broken Edges
After applying edge detection, object boundaries appear discontinuous.
(a) Interpret the scenario and explain why edges are broken.
(b) Identify key issues affecting boundary continuity.
(c) Apply a suitable method to improve edge connectivity.
(d) Justify your approach logically.
(e) Present your explanation clearly.

ANSWER:

**4. Broken Edges**

**(a) Interpret the scenario and explain why edges are broken**

After edge detection, some boundaries become discontinuous because of noise, weak gradients, or improper threshold settings.

**(b) Key issues**

- Noise.
- Weak edge responses.
- Incorrect threshold values.
- Missing edge pixels.

**(c) Suitable method to improve connectivity**

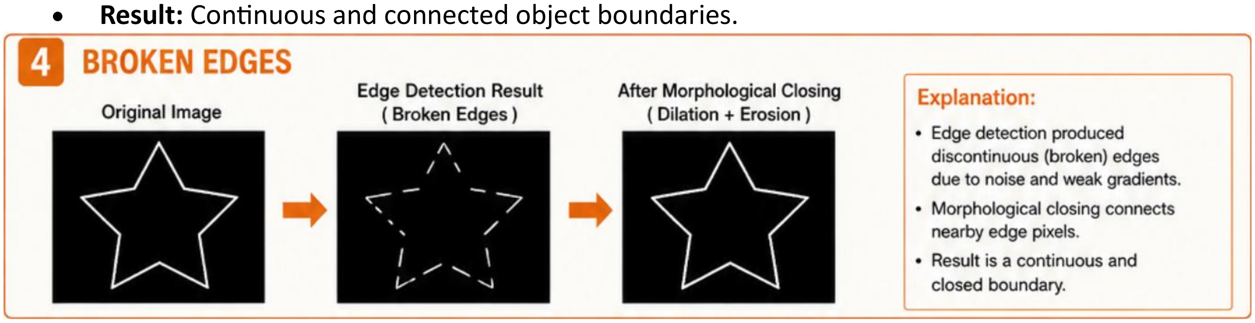
Morphological Closing (Dilation followed by Erosion)

**(d) Justification**

- Connects nearby edge segments.
- Fills small gaps in object boundaries.
- Preserves the overall object shape.
- Produces continuous edges for further processing.

**(e) Structured Explanation**

- **Scenario:** Discontinuous object boundaries.
- **Problem:** Broken edges.
- **Method:** Morphological Closing.
- **Result:** Continuous and connected object boundaries.



5. Combined Requirement
An industrial inspection system requires noise removal while preserving important edges for defect detection.
(a) Interpret the scenario and define the combined requirement.
(b) Identify key issues involving noise and edge preservation.
(c) Apply an appropriate sequence of techniques.
(d) Justify the order and selection with strong reasoning.
(e) Present your response in a clear and structured format.

ANSWER:

**5. Combined Requirement**

**(a) Interpret the scenario and define the combined requirement**

An industrial inspection system must remove image noise while preserving important edges so that defects such as cracks or scratches can be detected accurately.

**(b) Key issues**

- Noise may hide defects.
- Excessive smoothing may blur important edges.
- Edge preservation is essential for accurate inspection.

**(c) Appropriate sequence of techniques**

1. **Median Filter** (Noise Removal)
2. **Canny Edge Detection** (Edge Detection)

**(d) Justification**

- **Median Filter** removes impulse (salt-and-pepper) noise while preserving edges better than averaging filters.
- Applying **Canny Edge Detection** after noise removal produces accurate and continuous edges with fewer false detections.
- This sequence improves defect detection reliability.

**(e) Structured Explanation**

- **Scenario:** Industrial defect inspection.
- **Problem:** Remove noise without losing edges.
- **Technique Sequence:** Median Filter → Canny Edge Detection.
- **Result:** Noise-free image with sharp defect boundaries suitable for inspection.